LLM Theory Lab: Evidence Flow and the Cost of Structured Verification

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An empirical study of 172 completed runs across 24 finite tasks

Abstract

Research with language-model agents depends on the organization of evidence: who develops an argument, who examines the problem independently, when reports become visible, and how the final account incorporates them. LLM Theory Lab organizes these activities through a Builder, a Verifier who continues from independent reconstruction into disclosed review, and a fresh Director. We connect this architecture to an experiment with four workflows: an iterative single agent, an independent solver panel, the full Builder–Verifier–Director workflow, and an early-disclosure variant. This article analyzes the 172 completed runs from a 192-assignment study on 24 finite mathematics and computer-science tasks. Every workflow received four stages of 480 seconds with shared computation and submission interfaces. All 172 completed runs passed the existing Certified Task Success checker. Mean recorded active time was 12.02 minutes for the single agent, 11.17 for the panel, 18.38 for the full workflow, and 18.75 for early disclosure. On matched task-and-repeat comparisons with equal task weights, the full workflow used 54.0% more active time than the single agent and 65.8% more than the panel. The execution record also makes the architecture concrete: all 45 completed full-workflow runs preserved two sealed initial reports, resumed the same Verifier for review, and delivered both reports to review and synthesis. These findings give the theory an empirical form in which information separation, state continuity, evidence delivery, and resource consumption are jointly observable. The resulting account identifies the work performed by structured verification and its measured cost at a common finite-task success level.

1 Research organization as an experimental object

A research argument develops through several kinds of work. Someone proposes a route, someone tests its assumptions, and someone determines which conclusions the accumulated evidence supports. In a team of language-model agents, these activities can be assigned to distinct contexts and connected through explicit disclosure rules. The organization determines which evidence is available at each stage and which earlier work survives into the final submission.

The first draft of LLM Theory Lab proposed a Builder–Verifier–Director architecture around this idea [5]. The Builder develops and checks an argument. The Verifier first reconstructs the problem in a separate context, then examines the saved Builder report. The Director integrates the resulting evidence into a final account. Stable statements, explicit assumptions, preserved reports, and recorded

decisions provide the connective structure. This article develops that theory using the completed runs of the subsequent experiment.

The experiment gives the architecture two complementary empirical descriptions. Final certification records whether a workflow completed the fixed mathematical task with the required finite evidence. The execution record shows how that workflow organized the work: independent reports, continuation in the same context, disclosure events, artifact reads, and stage duration. In the 172 completed runs, final certification reached 100% in every condition. Work organization and resource consumption therefore supply the empirical distinctions within this population.

We study four conditions with shared resource ceilings. S is a strong single agent that continues across four stages with tools and private notes. P uses two independent solvers followed by a continuing Integrator. F implements the full Builder–Verifier–Director sequence with delayed initial disclosure. E retains the F roles and interfaces while exposing the Builder report at the Verifier's first stage. This design makes information flow a concrete experimental variable.

The paper contributes an operational account of the original theory, a descriptive comparison of completed finite-task solutions, and a trace-based analysis of report delivery. The central empirical pattern is clear: all four workflows reached the same certification level among completed runs, while F and E spent more active time in their four-stage workflows. F's two independently prepared reports and their subsequent delivery are visible throughout that sequence. The theory can consequently be expressed in terms of information separation, continuity, and the passage of evidence between research roles.

2 A theory of organized evidence

2.1 Information separation and continuity

Let a workflow policy π specify role assignments, visible information, tools, disclosure rules, and stopping conditions. For task x, model request M, and common resource ceilings b, the workflow produces a trajectory $\tau \sim \pi(M, x; b)$ and final artifact $a(\tau)$. The policy determines both the available information and the sequence in which agents act on it.

The key distinction in F is the Verifier's changing information set. Write X for the public task and allowed common materials, N_V for the Verifier's own evolving notes and native session state, and R_B, R_V for sealed Builder and Verifier reports. Once the requisite reports are available, the F policy has the form

$$\mathcal{I}^F_{V,2} = \{X, N_V\}, \tag{1}$$

$$\mathcal{I}^F_{V,3} = \{X, N_V, R_B, R_V\}. \tag{2}$$

The first stage preserves an independently prepared account. The second gives that same investigator access to a second account while retaining its own work. E changes the first information set to

$$\mathcal{I}^E_{V,2} = \{X, N_V, R_B\}, \tag{3}$$

when a valid Builder report is available. The recorded missing-report policy supplies explicit markers when the required report is absent.

This formulation locates independence at an observable boundary: the timing and availability of another role's material. Within-role continuity preserves accumulated calculations and notes. Across-role separation provides distinct starting contexts. Their combination gives the architecture a

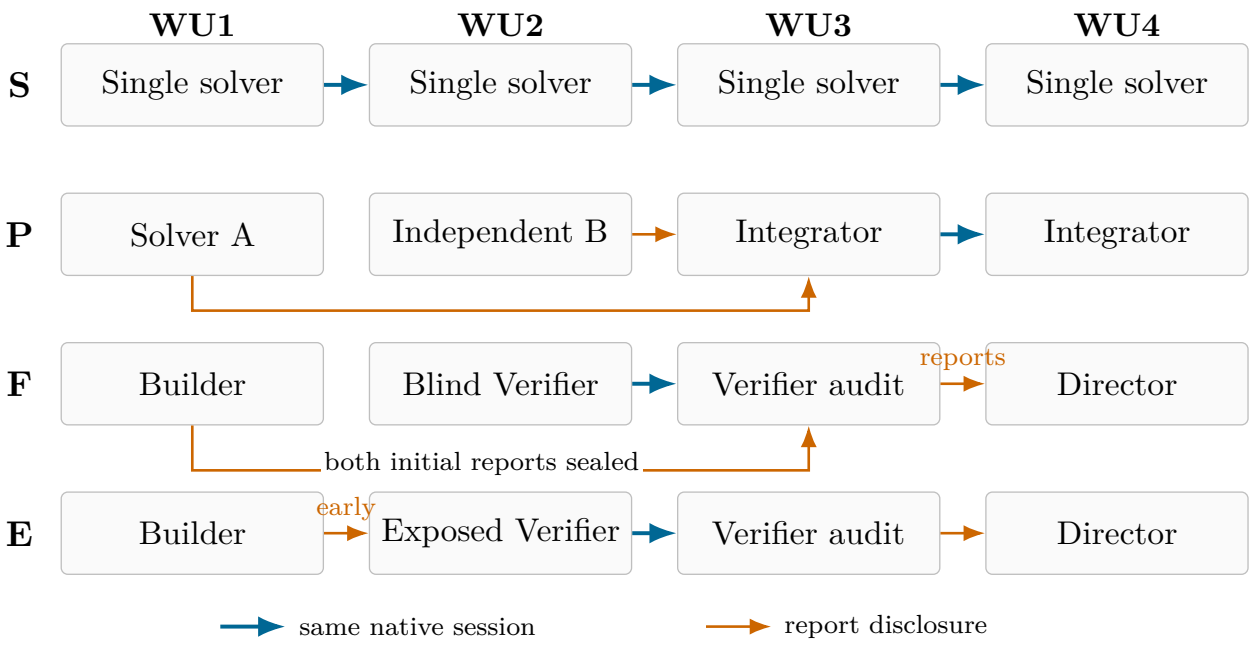


Figure 1: Four workflows under the same four-stage budget. Solid blue links denote continuation of the same native session. Orange arrows denote report disclosure. In F, both initial reports are sealed before disclosure to the continuing Verifier; the new Director receives the permitted initial reports and audit. In E, a valid Builder report is available at the Verifier's first stage. Independent roles and repeats begin with fresh state. Execution is serial across the experiment.

specific organization of inquiry: independent preparation followed by informed comparison. Figure 1 presents the four implemented paths.

2.2 From a claim to a shared research artifact

The original architecture attaches each claim to an exact statement, assumptions, dependencies, evidence, and a decision. This creates two related structures. A mathematical dependency structure records which claims support a conclusion. A disclosure structure records when a role gains access to the corresponding artifacts. The two structures meet when a reviewer examines a particular statement version and the Director cites the evidence used in the final account.

A useful unit for describing this process is

$$e = (c, v, \mathcal{A}, w, h, \delta), \tag{4}$$

where c is a claim identifier, v its version, $\mathcal{A}$ its assumptions, w the supporting argument or computational evidence, h an artifact identity, and δ its disclosure history. This notation makes explicit what a scientific handoff carries. The mathematical content is associated with a stable referent, and its transmission is associated with an observable event.

The execution record operationalizes the transmission part of this theory through sealed artifacts, content hashes, recipient allowlists, and actual read records. A delivered report can be represented by the conjunction

$$D(R, j) = \mathbf{1}\{R \text{ sealed}\}\,\mathbf{1}\{R \text{ disclosed to } j\}\,\mathbf{1}\{j \text{ read the matching artifact}\}. \tag{5}$$

Table 1: Implemented condition topology. A repeated role name denotes native session continuation. Every stage has a 480-second allowance.

Condition	WU1	WU2	WU3	WU4
S	Single solver	Single solver	Single solver	Single solver
P	Solver A	Independent Solver B	Integrator	Integrator
F	Builder	Independent Verifier	Verifier audit after double seal	Director
E	Builder	Verifier with early Builder report	Verifier audit	Director

Here D measures observed report delivery. The final task submission is evaluated by the finite-task checker; substantive adoption of an intermediate finding belongs to the claim-and-decision record. The present analysis measures completed task certification and the report-level delivery chain. Together, they establish a concrete path from private investigation to shared review and final submission.

2.3 Correction and preservation

The first draft distinguished correcting an initial error from losing an initially correct result during synthesis. For gradeable Builder and final artifacts, let $B, Y \in \{0, 1\}$ denote their respective success indicators. The identity

$$\mathbb{P}(Y = 1) - \mathbb{P}(B = 1) = \mathbb{P}(B = 0, Y = 1) - \mathbb{P}(B = 1, Y = 0) \tag{6}$$

expresses the net change across the later stages as corrections minus losses. This gives a theoretical reason to preserve the initial reports and the final artifact as distinct objects. It also links the independent Verifier's contribution to the Director's use of evidence: a useful finding enters the shared record through both discovery and subsequent adoption.

Equation 6 serves here as an interpretive decomposition. The empirical quantities reported below are final CTS, work consumption, and observed delivery. The preserved intermediate artifacts provide the corresponding units for a subsequent claim-level account of correction and retention. This connection keeps the theory tied to identifiable research objects at every stage.

3 Experimental realization

3.1 Tasks, conditions, and the analyzed population

The main task set contains 24 independently prepared finite tasks, four in each of six families: finite graphs, finite linear algebra, finite set systems, finite states and invariants, finite probability, and algorithm guarantees. Each family covers four combinations of mathematical truth and starting material: a true claim with useful incomplete material; a true claim with a misleading intermediate lemma; a false claim with helpful checking material; and a false claim accompanied by a plausible erroneous argument. The set contains 12 true and 12 false claims. Independent preparation records validate the domains, reference answers, and certificate interfaces.

The fixed queue assigns each task to S, P, F, and E twice, producing 192 assignments. Seed 20261003 determines two shuffled four-condition subblocks per task. Each independent repeat starts with fresh worker and session state. Table 1 summarizes the role topology.

This article analyzes the 172 selected attempts with final status `completed`: S45, P44, F45, and E38. Selection uses completion status and the batch's existing chronological attempt selection. CTS is then read from the saved grading result. The resulting accuracy and cost quantities describe completed executions. All 24 tasks appear for S, P, and F; E contributes 22 tasks. The full 192-assignment record remains the parent dataset. Two completed runs contain an intermediate-stage timeout followed by a sealed final submission; their stage outcomes remain part of this analysis.

3.2 Common resources and artifact interfaces

Every condition receives four work units (WU), each with a 480-second wall allowance, for a maximum of 1,920 stage seconds per run. The common computation allowance is six executions per WU and 24 per run. Each computation uses the existing nominal 15-second cumulative CPU budget with a 13-second early termination trigger. Stage allowances are fixed locally, and unused time remains with its original stage.

The recorded client request is `gpt-6-astra` with reasoning effort `ultra`, using the HTTP `openai_http` subscription route, Codex CLI 0.159.2, and sbx 0.46.0. These are request and client configuration fields. The three common tools are `read_allowed`, `write_note`, and `python_exec`. Logical artifact identifiers govern reads. Computation exports preserve the exact bytes of explicitly declared outputs before temporary-container cleanup, allowing certificates to enter the persistent submission record. The common evidence envelope, note limits, and report schema apply across conditions.

The same solver session continues through all four S stages; the P Integrator continues from WU3 to WU4; the F/E Verifier continues from WU2 to WU3. Private notes remain associated with their role. Cross-role inputs follow the saved disclosure rules. The evaluator receives the sealed batch after all assignments reach a terminal state. The run record binds attempts to execution revisions; append-only repairs addressed worker cleanup, transport diagnostics, computation startup, and broker activation while preserving task, role, model-request, scoring, and common resource settings.

3.3 Measurements and paired descriptions

Certified Task Success (CTS) is the existing conjunction

$$\mathrm{CTS} = \mathbf{1}\{\text{fixed goal correct}\}\,\mathbf{1}\{\text{required evidence valid}\}\,\mathbf{1}\{\text{scope preserved}\}. \tag{7}$$

For a proved finite claim, the checker verifies the required certificate and domain coverage. For a refuted claim, it checks an admissible witness against the stated premises and conclusion. The reported endpoint is finite-task certification; natural-language proof review is a separate evaluation layer.

Recorded active time is the sum of four native active intervals, from native launch to observed native end, with each interval capped at its stage deadline. It includes model and tool waits. Worker allocation, between-stage handoff, and post-native sealing and cleanup are recorded outside this measure. Thus active time measures the observed duration of the four solver stages. Computation counts are broker admission events. Original usage snapshots remain attached to their individual turns and sessions.

Condition summaries use all selected completed runs. Paired comparisons match the same task and repeat when both conditions completed. For comparison with condition a, let $\mathcal{R}_{i,a}$ be the available matched repeats for task i, and $\mathcal{T}_a$ the tasks with at least one such repeat. The reported time contrast is

$$\widehat{\Delta}_{F,a} = \frac{1}{|\mathcal{T}_a|} \sum_{i \in \mathcal{T}_a} \frac{1}{|\mathcal{R}_{i,a}|} \sum_{r \in \mathcal{R}_{i,a}} (T_{irF} - T_{ira}). \tag{8}$$

This gives each matched task equal weight after averaging its available repeats. Percent changes are ratios of the corresponding task-weighted mean times. The plots and tables are descriptive and reuse the saved completed-run analysis.

4 Certified completion and the distribution of work

4.1 A common certification level with different costs

All 172 completed runs pass CTS. The corresponding fractions are 45/45 for S, 44/44 for P, 45/45 for F, and 38/38 for E. This common certification level holds across all six task families represented in the completed population. Table 2 gives the resource profile accompanying these outcomes.

Table 2: Completed-run outcomes. Time is the sum of four recorded native active intervals. Computations are admitted tool executions.

	Completed	CTS	Mean min	Median min	Mean computations
S	45	45/45	12.02	11.44	5.07
P	44	44/44	11.17	10.51	5.64
F	45	45/45	18.38	17.63	7.93
E	38	38/38	18.75	18.04	7.71

S and P reached certified completion with shorter recorded active duration. F and E used more of the shared stage allowance, averaging about 18–19 active minutes per run. The mean computation counts show a similar division: approximately five to six admitted executions for S/P and eight for F/E. Each workflow used its common allowance in a characteristic way.

Figure 2 resolves this pattern across stages. In F, mean durations rise from 237 seconds for construction and 254 for independent verification to 310 for audit and 302 for synthesis. S averages 150, 189, 190, and 191 seconds; P averages 141, 136, 207, and 187. Thus the F–S/P difference extends across the workflow, with substantial time assigned to the review and final synthesis stages. These are observed stage durations, encompassing generation, interaction, and waits.

The saved computation-backend wall intervals average 4.79 seconds per S run, 5.18 for P, 7.71 for F, and 7.33 for E. Their recorded contribution is 0.65%–0.77% of active time. This locates most measured duration in the surrounding native-turn activity, which includes generation, reading, interaction, and waiting. It directs the cost analysis toward the full organization of solver work.

4.2 Matched tasks and repeated executions

The paired description preserves both task identity and repeat identity. F–S contains 43 matched run pairs across 24 tasks; F–P contains 42 across 24; F–E contains 38 across 22. Their task-weighted summaries appear in Table 3 and their task-level values in Figure 3.

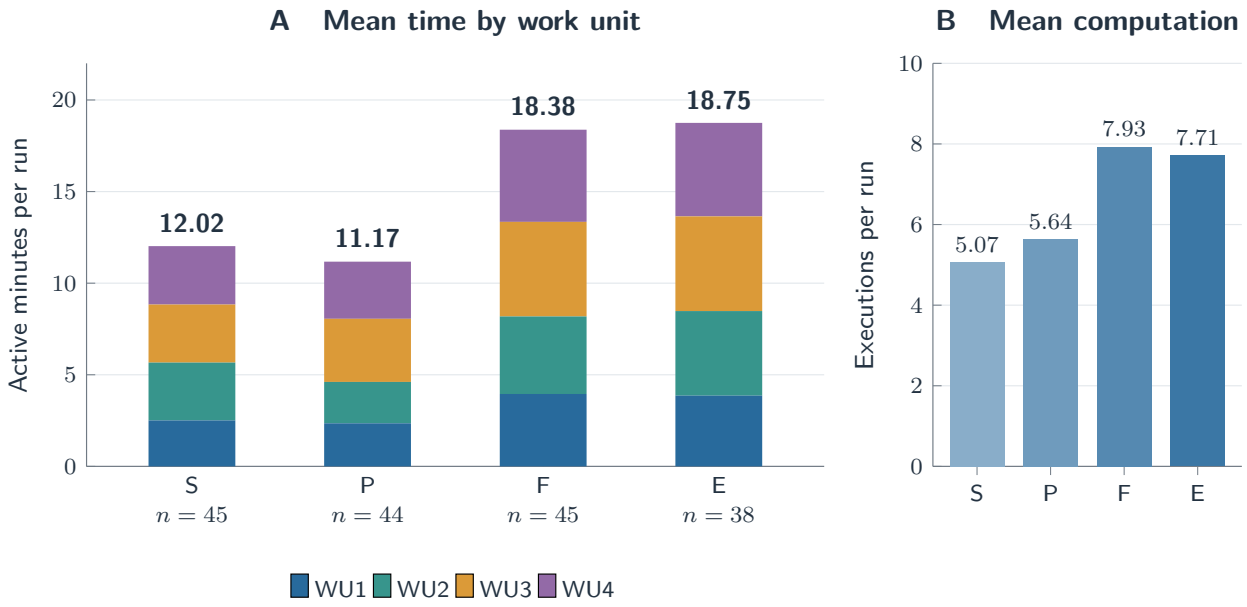


Figure 2: Recorded work in the 172 completed runs. Stacks show the mean active duration of each of the four stages. The companion panel shows mean admitted computation executions. Condition sample sizes are S45, P44, F45, and E38; all selected runs pass CTS. Every condition has the same four 480-second stage allowances and 24-execution run ceiling.

Table 3: Task-weighted paired active-time differences among completed runs. Available repeats are averaged within each task before tasks receive equal weight. The paired CTS difference is zero for all three comparisons.

Comparison	Run pairs	Tasks	Difference	Relative change	Tasks F slower
F–S	43	24	+6.46 min	+54.0%	24/24
F–P	42	24	+7.37 min	+65.8%	24/24
F–E	38	22	−0.64 min	−3.4%	6/22

F takes more active time than S and P on every one of the 24 matched task averages. The observed difference is therefore distributed across the task set. F–E is much smaller: F averages 38.56 fewer seconds, with shorter time on 16 of the 22 matched tasks. At the common completed-run CTS level, these comparisons describe the cost associated with the alternative organizations and the observed difference between their disclosure policies.

4.3 Six mathematical task families

Figure 4 displays mean active time by task family and condition. The completed sample retains the full six-family structure. S/P means range from about 10.3 to 14.0 minutes, while F/E means range from 17.2 to 20.8 minutes. Algorithm guarantees have the largest S/P means, and the cost of F remains visible across graphs, algebra, sets, states, probability, and guarantees. Each cell reports its contributing run count, preserving the different completed sample sizes.

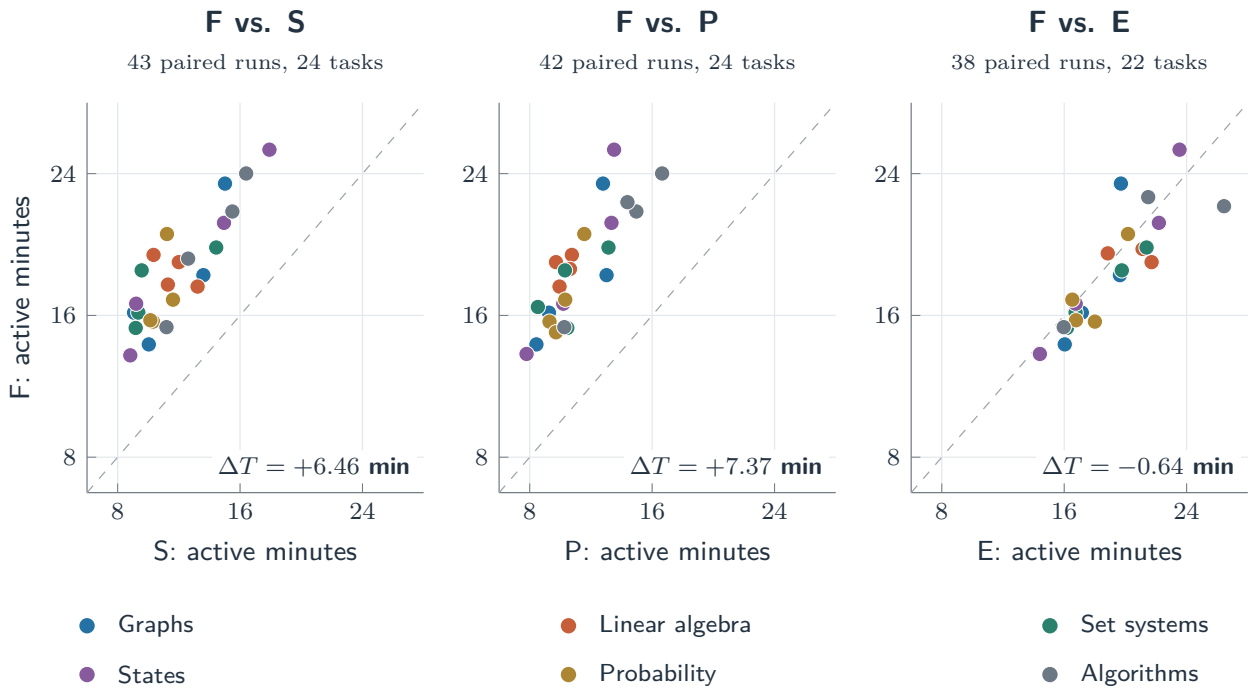


Figure 3: Task-level matched active-time comparisons. Each point represents one task after averaging its available matched repeats; the panels use the same task-and-repeat pairing as Equation 8. F is slower than S and P on every matched task. The F/E comparison is concentrated nearer equality. All points are drawn from the saved descriptive analysis of completed runs.

5 The observed evidence path

5.1 Independent preparation becomes disclosed review

The F record makes the intended transition from independent preparation to informed review observable in every completed run. All 45 F runs have a sealed Builder report, a sealed independent Verifier report, and a complete double-report checkpoint. F2's input catalogs and allocation receipts preserve the blind initial stage. F3 resumes the exact same native Verifier session and reads both initial reports. Each fresh Director also reads both reports. Table 4 summarizes these observations.

The initial reports consequently serve as preserved comparison objects. The same Verifier can relate its earlier account to the Builder's account, and the Director can examine both preparations alongside the audit. In 44 F runs, the audit was sealed and read by the Director. In the remaining run, the audit stage timed out and the Director completed a certified final submission from the permitted available material. The record keeps this distinction between initial reports, audit availability, and final success.

E realizes a different first review boundary. A Builder report was available and actually read during the initial Verifier stage in 37 completed runs. The remaining run carried the saved missing-report marker after a Builder-stage timeout. All 38 E Verifier sessions continued into audit, and all 38 audits sealed. Thus F and E provide concrete examples of delayed and early exposure while sharing role missions, stage budgets, and artifact interfaces.

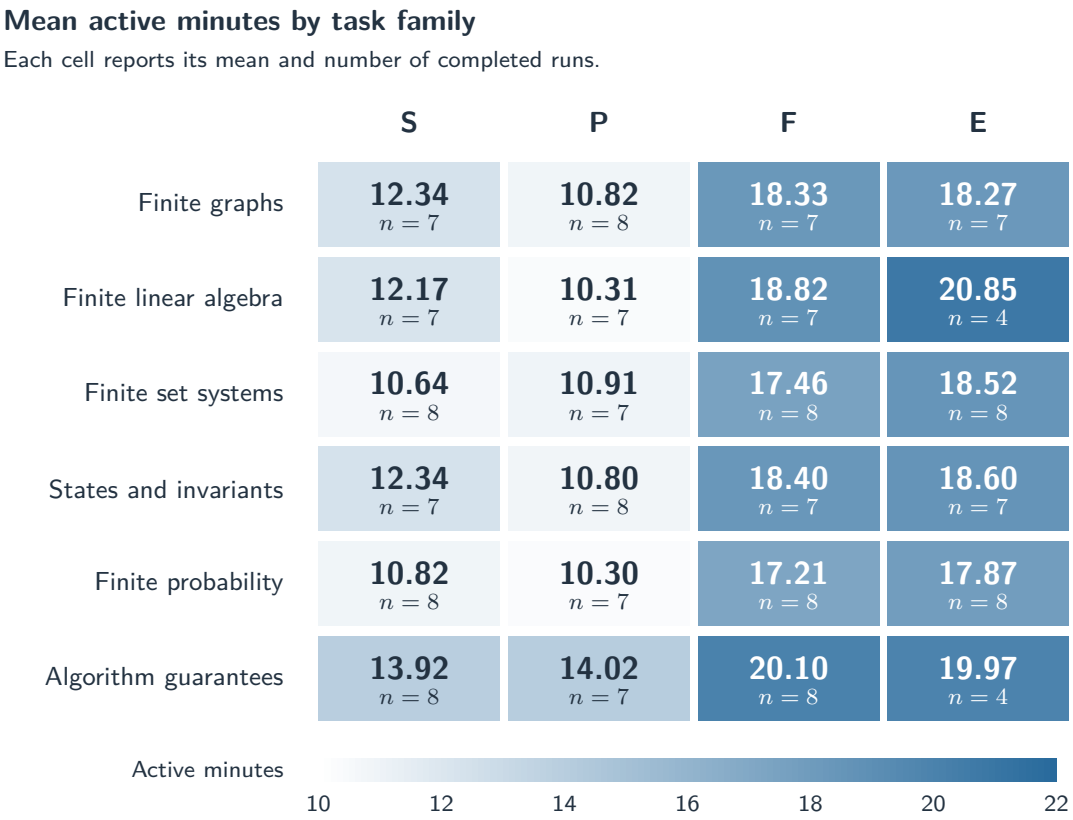


Figure 4: Mean active minutes by family and condition in the completed population. Each cell includes its run count. Every populated cell has 100% CTS. Values are descriptive family means; E contributes fewer completed tasks in the algorithm-guarantee and linear-algebra families.

5.2 Private continuity within separate roles

Across the 83 completed F/E runs, the record contains 249 distinct role workers and 249 native role sessions: one Builder, one Verifier, and one Director per run. Every Verifier retains its own session between the two stages. All 83 save private notes in WU2. Explicit reads of the corresponding notes appear in 43 F continuations and all 38 E continuations; the remaining two F cases retain the same native session.

These observations make separation and continuity complementary components of the method. Independent role identities establish distinct preparation contexts. Native continuation preserves the Verifier's accumulated work when the information set expands. Report disclosure provides the shared objects through which the separate contexts later communicate. The architecture therefore supports a defined movement from local investigation to collective examination.

5.3 Final success and intermediate work

The 172 completed runs contribute 688 stage records: 685 sealed stages, two intermediate timeouts, and one intermediate format-invalid submission. F's audit timeout and E's Builder timeout both lead to sealed, CTS-passing final submissions under the recorded missing-report rules. The format-invalid intermediate submission occurs in S's first stage, followed by successful later stages.

Table 4: Observed information flow in completed F/E runs. Counts refer to selected attempts. Early-disclosure report pairs retain their exposed preparation history.

Observable operation	F ($n = 45$)	E ($n = 38$)
Builder initial report sealed	45	37
Verifier initial report sealed	45	38
WU2 under blind independent policy	45	0
Early Builder report available and read in WU2	0	37
Complete initial-report pair	45	37
Exact native Verifier continuation WU2 $\rightarrow$ WU3	45	38
Separate Builder, Verifier, and Director identities	45	38
Verifier own notes saved in WU2	45	38
Explicit reads of matching own notes in WU3	43	38
Audit report sealed	44	38

The coexistence of these stage outcomes and completed final artifacts illustrates the value of preserving each stage separately. A trajectory contains the order in which materials became available, the state retained by a continuing role, and the evidence actually delivered to a later role. This is the empirical foundation for analyzing how an agent laboratory carries work forward.

6 What the experiment adds to the theory

6.1 A measurable architecture of research

The original theory assigned distinct responsibilities to construction, verification, and synthesis. The experiment gives those responsibilities an observable implementation. A blind input boundary marks independent preparation; a double seal preserves the initial accounts; a disclosure event changes the permitted information; native continuation carries the Verifier's state into review; artifact reads document delivery to the Director. These operations define the architecture at the level where research work actually passes between agents.

This moves the unit of analysis from a role label to a role's information and evidence history. The same "Verifier" label appears in F and E, while the information available at its first stage differs. The same Verifier identity appears in F2 and F3, while its access to another investigator's report changes. Together, these contrasts make disclosure timing and state continuity explicit parts of the research method.

6.2 An empirical cost for structured review

The completed-run results attach a quantitative cost to this organization. On matched tasks, F uses an additional 6.46 active minutes relative to S and 7.37 relative to P. All three reach the same measured certification level within the analyzed population. The extra recorded work accompanies a sequence that produces two initial accounts, a separate audit, and a final synthesis, with the initial accounts delivered to both later stages.

This yields a practical design question: what value does a research setting place on a preserved independent account and an explicit review history alongside a correct final artifact? For these finite tasks, S and P provide shorter paths to certified completion. F supplies a specific sequence of

independently prepared and subsequently shared artifacts at a measured additional cost. That cost is now a concrete quantity against which the scientific value of the intermediate work can be evaluated.

The F/E comparison further sharpens the role of disclosure. The two variants share the same role sequence and both use roughly 18–19 active minutes among completed runs. Their matched difference averages 39 seconds in favor of F. The protocol evidence shows the larger organizational distinction directly: F preserves 45 independently prepared report pairs before review; E gives the first Verifier stage early access to the Builder report when available. This supplies a precise experimental representation of the original independence policy.

6.3 Evidence preservation connects process to scientific value

Equation 6 places correction and retention inside the broader theory. The present results supply the final-success endpoint and the report delivery structure on which that decomposition operates. The preserved reports make it possible to identify a statement version, locate its initial evidence, follow its disclosure, and examine its treatment in the final account. This turns the architecture's concern with scientific reliability into a set of identifiable objects and transitions.

Three components emerge from the combined theory and experiment. Information separation determines how initial accounts are prepared. Continuity determines what an investigator carries into later work. Evidence delivery determines which artifacts reach review and synthesis. Final certification and observed resource use describe the outcome and cost of that organization. This gives LLM Theory Lab a concrete empirical vocabulary for studying agent research.

7 Relation to prior work

Self-consistency develops the value of multiple reasoning paths and answer aggregation [6]; multiagent debate studies repeated exchange among model instances [1]. The independent panel in this experiment provides a comparison with multiple attempts and later integration. F additionally specifies when a second account becomes visible and preserves a distinct initial Verifier report.

Self-Refine studies feedback and revision within a continuing model process [3]. The strong single-agent condition here likewise retains state, tools, and opportunities to revise across stages. F and P distribute parts of that work among fresh roles while retaining native continuation where the protocol requires it.

Work on test-time computation emphasizes allocation of inference effort [4]. The common stage ceilings and recorded durations give that allocation an observable form at workflow level. Cost-aware agent evaluation similarly motivates joint attention to successful outcomes and the resources used to obtain them [2]. This experiment connects that perspective to the preparation and transmission of mathematical evidence.

8 Conclusion

Across 172 completed executions on 24 finite tasks, all four workflows reached Certified Task Success. Their measured distinction was the amount and organization of work accompanying that success. The full LLM Theory Lab workflow used more active time than the single agent and independent panel on every matched task average. Its execution record also instantiated the intended sequence of blind reconstruction, preserved initial reports, disclosed review, and fresh synthesis.

The experiment advances the original theory by giving information separation, state continuity, and evidence delivery a measurable empirical form. It establishes a concrete resource cost for the implemented review sequence and a trace through which the movement of research artifacts can be examined. LLM Theory Lab can now be studied as an operating organization of research work, with completed solutions, explicit evidence paths, and recorded costs.

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A Stage measurements

Table 5: Mean recorded active seconds by stage in the completed population.

Condition	WU1	WU2	WU3	WU4	Sum
S	150.50	189.37	190.12	191.36	721.35
P	141.06	135.51	206.74	187.17	670.48
F	237.06	253.89	309.83	301.86	1102.65
E	231.56	277.07	310.07	306.30	1125.00

Table 6: Stage outcomes among the selected completed runs. Cells give sealed / timeout / format-invalid counts; each stage's denominator is its condition's completed-run count.

Condition	WU1	WU2	WU3	WU4
S	44/0/1	45/0/0	45/0/0	45/0/0
P	44/0/0	44/0/0	44/0/0	44/0/0
F	45/0/0	45/0/0	44/1/0	45/0/0
E	37/1/0	38/0/0	38/0/0	38/0/0

B Artifact provenance and reuse

This article develops the October 3 public revision of *Evaluation of LLM Theory Lab*, preserving its authorship, role architecture, and correction–retention decomposition. The empirical configuration follows the executed main batch and the clean-source binding. The new manuscript and figures are separate from the preserved first draft.

The parent experiment is batch `main_bc482c6515584a109cea9176e0af2c8c`. Its existing `RESULTS.json`, `STAGE_TIMING.json`, and `COMMON_SETTINGS.json` supply outcome, timing, and configuration records. The saved `completed172_v1/ANALYSIS.json` supplies the selected run list, condition and family summaries, and matched task descriptions. Its companion `MECHANISM.json` binds the report-delivery observations to 1,411 source metadata files. Figure data retain their source hashes and the exact saved values used for display.

The main task queue and candidate freeze are identified by SHA-256 prefixes `b7b1030e5f90` and `d4b04ce93f43`. The selected sample uses the batch's chronological selected attempts, including the recorded fresh infrastructure replacements. The analysis unit remains the assigned task and repeat. Two graph tasks share a template cluster in the parent study; the present figures report descriptive task values and paired task means.

All numerical results in this article reuse the saved completed-run analysis. The accompanying manuscript manifest records the first-draft PDF, authoritative architecture, analysis artifacts, runtime settings, figure sources, and final PDF. The editable LaTeX source and vector figures preserve the link from the article to its existing experimental evidence.